

# On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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## Abstract

We study  $G_n = \gcd(d_n a_n, d_n b_n)$  for the Apéry sequences  $(a_n), (b_n)$  with  $d_n = \text{lcm}(1, \dots, n)^3$ . We prove *unconditionally* that  $G_n = e^{o(n)}$  for a set of positive integers of natural density 1, with the quantitative bound  $\#\{n \leq N : \log G_n > \varepsilon n\} = O(N^{2/3})$  for every  $\varepsilon > 0$ . The key ingredient is a new bound  $Z(p) = O(p^{2/3})$  on the zero-count function  $Z(p) = \#\{j < p : p \mid b_j\}$ , derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes  $5 \leq p \leq 10^6$  gives  $Z(p) \leq 12$  with mean  $\approx 1$ . The pair count  $Z(p)/2$  fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ , so  $Z(p)$  is almost certainly unbounded. Under the weaker Cesàro hypothesis  $\sum_{p \leq x} Z(p) \ll \pi(x)$  (Hypothesis  $\bar{Z}$ ), which the Poisson model supports, we prove  $\log G_n = O(\sqrt{n})$  for density 1, matching the unconditional small-prime floor; the proof uses a dyadic divisor-incidence bound that reduces the leading-digit bottleneck from  $O(N^{3/2})$  to  $O(N^{4/3}/\log N)$ .

## 1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with  $(a_0, a_1) = (0, 6)$  and  $(b_0, b_1) = (1, 5)$ . Here  $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$  is the  $n$ -th Apéry number. Set  $d_n = \text{lcm}(1, \dots, n)^3$ ; Apéry [3] showed  $d_n a_n, d_n b_n \in \mathbb{Z}$ . Define  $G_n = \gcd(d_n a_n, d_n b_n)$ . Write  $P(n) = 34n^3 + 51n^2 + 27n + 5$  for the middle coefficient.

## 2 The Wronskian identity

**Lemma 1.** *For all  $n \geq 1$ ,  $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ .*

*Proof.* The Casorati determinant satisfies  $W_{n+1}/W_n = n^3/(n+1)^3$  from (1). Since  $W_1 = a_1 b_0 - a_0 b_1 = 6$ , telescoping gives  $W_n = 6/n^3$ .  $\square$

**Corollary 2.** *For every prime  $p \geq 5$ ,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

*Proof.*  $G_n$  divides  $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$ . For  $p \geq 5$ ,  $v_p(6) = 0$  and  $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$ .  $\square$

### 3 Small primes: unconditional bound

**Proposition 3.**  $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n}).$

*Proof.* By (2),  $v_p(G_n) \log p \leq 3 \log n$  for every prime  $p \geq 5$ . Primes  $p = 2, 3$  contribute at most  $O(\log n)$  each. There are  $O(\sqrt{n}/\log n)$  primes up to  $\sqrt{n}$ , so the sum is at most  $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$ .  $\square$

### 4 The multiplicative Lucas congruence

**Theorem 4** (Gessel [9]). *For every prime  $p$  and  $n = \sum_{i=0}^s n_i p^i$  in base  $p$ ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$  and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [13] and McIntosh [12].

### 5 The zero-count function

**Definition 5.** For a prime  $p \geq 5$ , define  $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$ .

#### 5.1 Unconditional bound

**Lemma 6** (No consecutive zeros). *For every prime  $p \geq 5$  and every  $j \in \{0, \dots, p-2\}$ ,  $b_j$  and  $b_{j+1}$  cannot both vanish modulo  $p$ .*

*Proof.* Suppose  $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$  with  $0 \leq m \leq p-2$ . The recurrence (1) at  $n = m$  gives  $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$ , so  $m^3 b_{m-1} \equiv 0 \pmod{p}$ . Since  $1 \leq m \leq p-2$  implies  $p \nmid m$ , we get  $b_{m-1} \equiv 0$ . Iterating gives  $b_0 \equiv 0 \pmod{p}$ , contradicting  $b_0 = 1$ .  $\square$

**Lemma 7** (Gap polynomial). *Fix  $h \geq 2$ . Suppose  $b_m \equiv 0 \pmod{p}$  and  $m+h \leq p-1$ . Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \quad (4)$$

where  $C_h(m)$  is a rational function of  $m$  whose numerator (after clearing the denominator  $\prod_{j=1}^{h-1} (m+j+1)^3$ ) is a polynomial of degree exactly  $3(h-1)$  with positive leading coefficient.

In particular, if  $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$ , then by Lemma 6  $b_{m+1} \not\equiv 0$ , so  $C_h(m) \equiv 0 \pmod{p}$ , and  $m$  is constrained to at most  $3(h-1)$  residue classes modulo  $p$ , provided  $C_h$  is not identically zero over  $\mathbb{F}_p$ .

*Proof.* We iterate the recurrence from the initial condition  $(b_m, b_{m+1}) = (0, b_{m+1})$ . At step  $n = m+1$ :  $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$ , since  $b_m = 0$  kills the second term. At step  $n = m+j$  for  $j \geq 2$ :  $b_{m+j+1}$  is a  $\mathbb{Q}(m)$ -linear combination of  $b_{m+j}$  and  $b_{m+j-1}$ , both already rational multiples of  $b_{m+1}$ . The leading coefficient of  $P(n)$  is 34 and the subtracted term has leading coefficient 1, so at each step the degree in  $m$  increases by 3 and the leading coefficient remains positive. After  $h-1$  steps the numerator has degree  $3(h-1)$ .

For  $h = 2$ :  $C_2(m) = P(m+1)/(m+2)^3$ , with numerator  $P(m+1) = (2m+3)(17m^2+51m+39)$ , a cubic with leading coefficient 34.  $\square$

**Lemma 8** (Nonvanishing over  $\mathbb{F}_p$ ). *For every prime  $p \geq 7$  and every  $2 \leq h \leq p$ , the numerator of  $C_h$  is not the zero polynomial modulo  $p$ .*

*Proof.* Write  $N_h$  for the numerator of  $C_h$  (a polynomial of degree  $3(h-1)$ ), satisfying  $N_{h+1}(m) = P(m+h)N_h(m) - (m+h)^6 N_{h-1}(m)$ , with  $N_1 = 1$  and  $N_2 = P(m+1)$ .

*First evaluation.* Equation (4) at  $m = -1$  (where  $b_{-1} = 0$ ,  $b_0 = 1$ ) gives  $C_h(-1) = b_{h-1}$ , hence  $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$ .

*Second evaluation.* Setting  $m = -2$  in the recurrence gives  $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$ , with  $N_1(-2) = 1$  and  $N_2(-2) = P(-1) = -5$ . By induction using the Apéry recurrence at  $n = h-2$ :

$$N_h(-2) = -5b_{h-2}((h-2)!)^3 \quad (h \geq 2). \quad (5)$$

*Conclusion.* If  $N_h \equiv 0 \pmod{p}$  as a polynomial ( $2 \leq h \leq p$ ,  $p \geq 7$ ), then  $N_h(-1) \equiv N_h(-2) \equiv 0$ , forcing  $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$  (since  $p \nmid 5$  and  $p \nmid (h-1)!, (h-2)!$ ). These are consecutive zeros, contradicting Lemma 6.  $\square$

**Proposition 9.** *For every prime  $p \geq 7$ ,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (6)$$

where  $H$  is any integer with  $2 \leq H \leq p$ . Optimizing  $H = \lfloor (p/3)^{1/3} \rfloor + 1$  gives  $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$  unconditionally for all primes. In particular,  $Z(p) = o(p)$ .

*Proof.* Partition  $\{0, \dots, p-1\}$  into  $\lceil p/H \rceil$  consecutive blocks of length at most  $H$ . In each block, mark the first zero; all other zeros have gap  $h \in \{2, \dots, H-1\}$  to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap- $h$  pair constrains  $m$  to at most  $3(h-1)$  residue classes mod  $p$  (the denominator factors  $(m+j+1)^3$  are nonzero since  $0 \leq m+j+1 \leq p-1$ ). By Lemma 8,  $C_h \not\equiv 0$  over  $\mathbb{F}_p$ , so the numerator roots are at most  $3(h-1)$ . Summing:

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real  $H > 0$  occurs at  $H = (p/3)^{1/3}$ , giving the bound. For  $p = 2, 3, 5$ :  $Z(p) = 0$  (since  $b_j$  for  $j < 5$  is coprime to  $2, 3, 5$ ).  $\square$

*Remark 10.* The leading coefficient of the numerator of  $C_h$  is  $U_{h-1}(17)$ , the Chebyshev polynomial of the second kind evaluated at 17, satisfying  $\ell_1 = 1$ ,  $\ell_{h+1} = 34\ell_h - \ell_{h-1}$ . Let  $N_h(m)$  denote the cleared numerator of  $C_h$ . The identity  $P(-u-1) = -P(u)$  (evident from  $P(n) = (2n+1)(17n^2 + 17n + 5)$ ) propagates through the recurrence to give  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ , verified computationally for  $h \leq 200$ . For even  $h$ ,  $N_h$  is an *odd* polynomial in  $(m + (h+1)/2)$ , forcing the structural factor  $(2m+h+1)$ . At  $h = 2$ : factor  $(2m+3)$  in  $P(m+1) = (2m+3)(17m^2 + 51m + 39)$ . The computational data (Section 9) suggests  $Z(p) = O(1)$ , far stronger than  $O(p^{2/3})$ .

*Remark 11* (Content restriction). A stronger nonvanishing holds: if  $p \geq 7$  divides the content of  $N_h$  (the gcd of all its coefficients), then  $h \geq 2p$ . Indeed,  $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$  for  $1 \leq r \leq h$  (by induction from the Apéry recurrence); choosing  $r = 1, 2$  and using  $p \nmid (h-1)!, (h-2)!, 5$  for  $h \leq p$  recovers Lemma 8, while the range  $p < h < 2p$  is excluded similarly via  $r = h-p+1$ . Setting  $\beta_n = (n!)^3 b_n$  and  $F(z) = \sum_{n \geq 0} \beta_n z^n$ , the endpoint factorization packages into the double generating function  $\sum_{r,s \geq 0} \frac{\beta_{r+s+1} (-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u) F(v)$ , showing that the boundary array is a rank-one tensor.

*Remark 12* (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$|\text{Res}(N_h, N_{h+1})| = \prod_{j=1}^{h-1} (j!^3 b_j)^6. \quad (7)$$

This follows from the recursion  $\text{Res}(N_h, N_{h+1}) = N_h(-h)^6 \text{Res}(N_{h-1}, N_h)$  (since  $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$ ), combined with  $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1} ((h-1)!)^3 b_{h-1}$  (the reflection identity and the first evaluation from Lemma 8). Verified for  $h = 2, 3, 4, 5$ ; the cases  $h = 2, 3, 4$  give  $\text{Res}(N_2, N_3) = -5^6$ ,  $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$ ,  $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$ .

A key consequence:  $p \mid \text{Res}(N_h, N_{h+1})$  if and only if  $p \mid b_j$  for some  $1 \leq j \leq h-1$  (when  $p > h$ , so factorials are coprime to  $p$ ). In particular,  $N_2$  and  $N_3$  have disjoint root sets modulo  $p$  for all  $p \geq 7$  (since  $5^6 \neq 0$ ).

*Remark 13* (Sharpness of the bound). The exponent  $2/3$  in Proposition 9 is optimal for any argument using only the degree bounds  $\deg N_h = 3(h-1)$ . The exact extremal problem  $\max\{\sum r_h : r_h \leq 3(h-1), \sum h r_h \leq p\}$  has value  $\frac{3}{2} p^{2/3} + O(p^{1/3})$ , with leading constant  $3/2$  achieved by filling gap sizes  $h = 2, 3, \dots, H$  to capacity and using the remaining space at  $h = H+1$ , where  $H \sim p^{1/3}$ . Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield  $O(p^{1/2})$ : an abstract polynomial model on  $\sim p^{2/3}$  points can satisfy  $\deg F_h \leq 3(h-1)$  for all relevant  $h$ . Improving the exponent requires genuinely arithmetic input about the family  $(N_h)$ .

*Remark 14* (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll’s condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m) N_h(m+1) - N_{h+1}(m) N_{h+1}(m+1) = - \prod_{j=2}^{h+1} (m+j)^6. \quad (8)$$

This is an identity in  $\mathbb{Z}[m]$ , verified for  $h \leq 7$ . A consequence: for  $m \not\equiv -j \pmod{p}$  ( $2 \leq j \leq h+1$ ), the pairs  $(m, h+2)$  and  $(m+1, h)$  cannot both lie in the zero relation  $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$ , since  $N_{h+1}(m) N_{h+1}(m+1)$  would equal a nonzero product of sixth powers. After a meromorphic gauge  $\tau_h(x) = N_h(x)/d_h(x)$  satisfying  $d_{h+2}(x) d_h(x+1) = \prod_{j=2}^{h+1} (x+j)^6$ , this becomes an exact  $SL_2$ -tiling:  $\tau_{h+1}(x) \tau_{h+1}(x+1) - \tau_{h+2}(x) \tau_h(x+1) = 1$ , equivalently an  $A_1$   $T$ -system (discrete Liouville equation). A zero of  $\tau$  forces its transverse neighbors to be units (*singularity confinement*), but abstract  $SL_2$ -tilings can have zeros of density  $1/2$  (e.g.,  $q_n = 0, 1, 0, -1, \dots$  with  $q_n^2 - q_{n-1} q_{n+1} = 1$ ), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 20), and polynomial degree bounds—is essential.

*Remark 15* (Projective orbit reformulation). Let  $B_n = n!^3 b_n$  and  $D_n$  be the second solution of the renormalized recurrence  $w_{n+1} = P(n) w_n - n^6 w_{n-1}$  with  $(D_0, D_1) = (0, 1)$ . Their Casoratian is  $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$ , nonzero for  $n < p$ . Define the *projective Apéry orbit*  $\pi: \{0, \dots, p-1\} \rightarrow \mathbb{P}^1(\mathbb{F}_p)$  by  $\pi_n = [B_n : D_n]$ . For  $0 \leq m < m+h < p$ ,  $N_h(m) \equiv 0 \pmod{p}$  if and only if  $\pi_m = \pi_{m+h}$ . Thus the zero-count  $Z(p)$  equals the number of distinct collisions in the orbit  $\pi$ . The reflection symmetry  $P(-1-n) = -P(n)$  implies  $b_{p-1-n} \equiv b_n \pmod{p}$ , making the orbit a palindrome:  $\pi_n = \pi_{p-1-n}$ . In particular,  $Z(p)$  is even except at the non-ordinary primes  $p \mid a_p(f)$  (at most two primes  $p \leq 10^5$ ).

*Remark 16* (Galois structure). For each fixed  $h$ , the Galois group of the splitting field of  $N_h$  over  $\mathbb{Q}$  is not the full symmetric group: the reflection  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$  forces roots into pairs  $\{r, -r-h-1\}$ , and  $\text{Gal}(N_h) \cong C_2 \wr S_m$  (hyperoctahedral group) with  $m = \lfloor 3(h-1)/2 \rfloor$  (confirmed exactly for  $h \leq 5$ ,  $2 \leq h \leq 9$  by obstruction tests). By the Chebotarev density theorem, the average number of  $\mathbb{F}_p$ -roots of  $N_h$  is 1 for odd  $h$  and 2 for even  $h$ . This prediction is verified: for 262 primes in  $[1000, 3000]$ , the measured averages are 0.98, 2.03, 1.11, 2.07, 1.11, 1.88 for  $h = 3, \dots, 8$ .

**Proposition 17** (Hyperoctahedral Poisson law). *Assume  $\text{Gal}(N_h) \cong C_2 \wr S_m$  with  $m = \lfloor 3(h-1)/2 \rfloor$ . Let  $Y_h(p)$  be the number of non-central reflection-pairs of roots of  $N_h$  that split completely over  $\mathbb{F}_p$ . Then for each fixed  $k \leq m$ , the factorial moments in the prime-aspect satisfy*

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where  $(Y)_k = Y(Y-1) \cdots (Y-k+1)$ . Since these match the factorial moments of  $\text{Poisson}(1/2)$  for all  $k$ , the prime-aspect distribution of  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$  as  $h \rightarrow \infty$ .

*Proof.* In  $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m, \varepsilon \in \{\pm 1\}^m\}$ , a positive fixed point of  $g = (\sigma, \varepsilon)$  is an index  $i$  with  $\sigma(i) = i$  and  $\varepsilon_i = +1$ . The Frobenius at  $p$  acts on the  $m$  reflection-pairs, and  $Y_h(p)$  equals the number of positive fixed points. For any  $k$  distinct indices  $i_1, \dots, i_k$ , the fraction of  $g \in G$  fixing all  $k$  positively is

$$\frac{2^{m-k} (m-k)!}{2^m m!} = \frac{1}{2^k (m)_k}.$$

Summing over all  $(m)_k$  ordered  $k$ -tuples,  $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$ , which matches the factorial moments of  $\text{Poisson}(1/2)$ . The Chebotarev density theorem then gives the prime-aspect convergence (Remark 19 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all  $k \leq m$ ,  $Y_h(p) \rightarrow \text{Poisson}(1/2)$  as  $m \rightarrow \infty$ .  $\square$

*Remark 18* (Prime-aspect vs. orbit-aspect). Proposition 17 is a *prime-aspect* statement: for fixed  $h$  and varying  $p$ , the root-pair count  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$ . The zero-count  $Z(p)$  aggregates contributions from all gaps  $h = 1, \dots, p-1$  at a *fixed* prime  $p$ , so the orbit-aspect independence needed to deduce  $Z(p)/2 \sim \text{Poisson}(1/2)$  is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

*Remark 19* (GRH-effective error). For each fixed  $k$ , the  $k$ -th factorial moment does not require the full splitting field  $K_h$  (of degree  $2^m m!$ , double-exponential in  $h$ ). Let  $E_{h,k}$  be the fixed field of the stabilizer of an ordered signed  $k$ -tuple with distinct underlying reflection pairs; then  $[E_{h,k} : \mathbb{Q}] = 2^k (m)_k = O_k(h^k)$ . Under GRH for  $\zeta_{E_{h,k}}$ ,

$$\frac{1}{\pi(x)} \sum_{\substack{p \leq x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where  $D_{h,k} = |\text{Disc}(E_{h,k})|$ . The Apéry recurrence gives the coefficient-height bound  $\log \|N_h\|_1 = O(h \log h)$ , from which  $\log D_{h,k} \ll_k h^{k+2} \log h$ . Thus an  $O(h^{-C})$  error holds in the polynomial range  $x \geq h^{2k+4+2C+\varepsilon}$ , for each fixed  $k$ .

At  $x = 10^6$  with  $h = 10$  ( $m = 13$ ), the degree contributions alone for  $k = 1, 2, 3$  are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

## 5.2 Dual bounds: the restart identity and column estimate

The bound  $Z(p) = O(p^{2/3})$  controls zeros along a *fixed row* (varying  $m$  for a given gap  $h$ ). A dual bound controls zeros along a *fixed column* (varying  $h$  for a given starting point  $x_0$ ).

**Lemma 20** (Restart identity). *Suppose  $N_r(x_0) = 0$  and  $N_{r+1}(x_0) \neq 0$  for some  $x_0 \in \mathbb{F}_p$  and  $r \geq 1$ . Then for every  $d \geq 0$ ,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \quad (9)$$

*In particular, if  $N_{r+d}(x_0) = 0$  as well, then  $N_d(x_0 + r) = 0$ .*

*Proof.* The sequence  $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$  has  $Y_0 = 0$ ,  $Y_1 = 1$ , and satisfies  $Y_{d+1} = P((x_0+r)+d) Y_d - ((x_0+r)+d)^6 Y_{d-1}$ , which is the defining recurrence for  $N_d(x_0 + r)$ .  $\square$

**Proposition 21** (Column bound). *For every  $x_0 \in \mathbb{F}_p$  and every interval  $I$  of  $H$  consecutive  $h$ -values with  $H \leq p$ ,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \frac{3^{4/3}}{2} H^{2/3} + O(H^{1/3} + 1). \quad (10)$$

*Proof.* Split  $I$  at the unique transition  $h$  where  $x_0 + h \equiv 0 \pmod{p}$  (if it exists in  $I$ ), creating at most two nonsingular pieces plus an  $O(1)$  additive cost.

On a nonsingular piece, order the zero levels. For threshold  $L \geq 2$ : a gap of length  $d \leq L$  between consecutive zeros at levels  $r < r + d$  gives  $N_d(x_0 + r) = 0$  by Lemma 20. For fixed  $d$ , there are at most  $3(d-1)$  values of  $x_0 + r$  satisfying this (by Lemma 8 and  $\deg N_d = 3(d-1)$ ). Gaps longer than  $L$  contribute at most  $H/(L+1)$  zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^L 3(d-1) + O(1) = \frac{H}{L+1} + \frac{3}{2}L(L-1) + O(1).$$

Optimizing  $L \asymp (H/3)^{1/3}$  gives the stated bound.  $\square$

Computationally, the column count at mesoscopic scale  $H = \lfloor \sqrt{p} \rfloor$  is far below the  $O(H^{2/3})$  bound: for all primes  $p \leq 100,003$  and all  $x_0$ ,  $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$ . For  $p \geq 10,007$ , the maximum is  $\leq 1$ .

*Remark 22* (Rectangular incidence estimate). The row bound  $|R_h| \leq 3(h-1)$  and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p, 2 \leq h \leq H, N_h(x) \equiv 0\} \leq \min\{H^2, pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at  $H \sim p^{3/4}$ . Note that a uniform bound  $R_p(H) \leq CH$  for all primes is *false*: by the Chebotarev density theorem, for each fixed  $H$ , infinitely many primes  $p > H^2$  have every  $N_h$  ( $2 \leq h \leq H$ ) splitting completely over  $\mathbb{F}_p$ , giving  $R_p(H) = \frac{3}{2}H(H-1)$ . The correct target is the *mesoscopic* estimate  $R_p(\lfloor \sqrt{p} \rfloor) = O(\sqrt{p})$ , where the split-prime obstruction does not apply because  $H = \sqrt{p}$  grows with  $p$ . More generally, if  $R_p(H) \ll H^\beta$  then  $Z(p) \leq p/H + R_p(H)$  is optimized at  $H \sim p^{1/(\beta+1)}$ , giving  $Z(p) \ll p^{\beta/(\beta+1)}$ : the current  $\beta = 2$  gives  $2/3$  and  $\beta = 1$  yields  $1/2$ . Reflection forces  $R_p(H) \geq H/2$  from central roots, so  $\beta = 1$  is the intrinsic limit of this first-moment method.

### 5.3 The continuant Bezout identity

Write  $S_d(x) = \prod_{j=2}^d (x + j)$ .

**Proposition 23** (Bezout identity). *For integers  $2 \leq d < e$ ,*

$$N_{e-1}(x+1) N_d(x) - N_{d-1}(x+1) N_e(x) = S_d(x)^6 N_{e-d}(x+d). \quad (11)$$

*In particular, after saturating the cut-edge divisor  $S_d$ ,*

$$(N_d, N_e) : S_d^\infty = (N_d, N_{e-d}(x+d)) : S_d^\infty.$$

*Proof.* The continuant addition formula (splitting the interval  $[1, e-1]$  at position  $d$ ) gives  $N_e = N_d N_{e-d+1}(x+d-1) - (x+d)^6 N_{d-1} N_{e-d}(x+d)$ . The Dodgson identity at index  $d-2$  gives  $N_{d-1}(x+1) N_{d-1} - N_{d-2}(x+1) N_d = \prod_{j=2}^{d-1} (x+j)^6$ . Combining these two identities yields (11). Verified computationally for all  $2 \leq d < e \leq 9$ .  $\square$

**Corollary 24** (Separated-block criterion). *Away from the cut-edge divisor  $S_d(x) = 0$ ,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod{p} \iff N_d(\alpha) \equiv N_{e-d}(\alpha + d) \equiv 0 \pmod{p}.$$

*In particular,  $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \leq 3(\min(d, e-d) - 1)$ , which improves the naive  $3(d-1)$  when  $e - d < d$ .*

*Remark 25* (Separated-block resultant). Define  $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$ . The Bezout identity gives

$$|\text{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} ((j!)^3 b_j)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ( $e = d + 1$ ),  $N_1 = 1$  gives  $\mathcal{R}_{d,1} = 1$ , recovering the adjacent resultant formula. For nonadjacent indices,  $\mathcal{R}_{d,r}$  is genuinely new. The reflection identity gives  $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$ . The subtraction step does *not iterate*: after one application,  $N_d(x)$  and  $N_{e-d}(x+d)$  live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For  $d = 2$ , writing  $P(y) = (2y+1)Q(y)$  with  $Q(y) = 17y^2 + 17y + 5$  and  $\theta$  a root of  $Q$ , the norm decomposition  $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)} N_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$  (where  $G_r = N_r(x+1)$ ) gives  $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$ . At  $p = 71$ :  $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$  with  $-5/2 \notin \{-1, -2, -3, -4\}$ , a nonboundary common root inside the mesoscopic range ( $4^3 < 71$ ). This shows the boundary-only gcd statement is *false*.

*Remark 26* (Collision energy and the  $H^{1/2}$  barrier). The column exponent  $2/3$  is the natural barrier for the consecutive-gap argument: even if each gap of length  $d$  has only  $d$  (rather than  $3(d-1)$ ) admissible starts, the sum  $\sum_{d \leq L} d \asymp L^2$  still gives  $H/L + L^2$ , optimized at  $H^{2/3}$ . To achieve  $H^{1/2}$ , one needs the aggregate collision energy  $\mathcal{E}_p(H) = \sum_{d+r \leq H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$ , which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The  $A_1$  T-system  $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$  admits tame solutions with zero density  $1/2$ , e.g.  $\tau_{2k} = 0$ ,  $\tau_{2k+1} = (-1)^k$ , so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing  $M(n) = \begin{pmatrix} P(n) & -n^6 \\ 1 & 0 \end{pmatrix}$ , so that  $N_h(x) = 0$  iff a fixed matrix coefficient of the interval product  $M(x+h-1) \cdots M(x+1)$  vanishes), the conjectured  $\mathcal{E}_p(H) \ll H^{3/2}$  is analogous to a Rudnev-type point-plane incidence bound in  $\text{PGL}_2(\mathbb{F}_p)$ , subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

| $p$                            | 71   | 503  | 2003 | 5003 | 7919 | 20011 | 50021 | $10^5+3$ |
|--------------------------------|------|------|------|------|------|-------|-------|----------|
| $H = \lfloor \sqrt{p} \rfloor$ | 8    | 22   | 44   | 70   | 88   | 141   | 223   | 316      |
| $R_p(H)/H$                     | 1.75 | 1.59 | 1.36 | 1.50 | 1.75 | 1.63  | 1.56  | 1.46     |
| $\mathcal{E}_p(H)$             | 1    | 2    | 2    | 2    | 10   | 0     | 2     | 0        |
| $\mathcal{E}_p(H)/H^{3/2}$     | .044 | .019 | .007 | .003 | .012 | 0     | .001  | 0        |

The ratio  $R_p(H)/H$  clusters around  $3/2$  (the Chebotarev prediction from Remark 16), and  $\mathcal{E}_p(H)/H^{3/2} \rightarrow 0$ , consistent with  $\mathcal{E}_p(H) \ll H^{3/2}$ . The key arithmetic object is the shifted resultant  $S_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$ : for  $p \nmid S_{d,r}$ , the gap polynomials  $N_d$  and  $N_r(\cdot+d)$  share no root over  $\mathbb{F}_p$ , so that gap pair contributes zero to  $\mathcal{E}_p$ . A Hadamard bound gives  $\log |S_{d,r}| \ll (d+r)^2 \log(d+r)$ , so

the total number of prime divisors across all pairs  $(d, r)$  with  $d + r \leq H$  is  $O(H^4 \log H)$ . Averaging over the  $\sim X/\log X$  primes in  $(X, 2X]$  gives  $\mathcal{E}_p(H) \ll H^{1/2}$  for all but  $o(X/\log X)$  primes, provided  $H \leq X^{2/7-\varepsilon}$ —rigorous but far from the global target  $H = p - 1$ .

A sharper route uses the bipartite incidence graph  $\Gamma_p(H)$ : left vertices are gap lengths  $d$ , right vertices are gap lengths  $r$ , with an edge when  $N_d$  and  $N_r(\cdot + d)$  share a root over  $\mathbb{F}_p$ . Each edge certifies a triple collision, so  $\mathcal{E}_p(H) = e(\Gamma_p(H))$ . A uniform fiber bound  $\deg_L(d) \leq C$  (the number of  $r$ -partners for each  $d$ ) and a uniform codegree bound (at most  $C$  common right neighbors for any two left vertices) would place  $\Gamma_p(H)$  in the regime of the Kővári–Sós–Turán theorem, giving  $\mathcal{E}_p(H) \ll H^{3/2}$ —the collision-energy scale needed for the mesoscopic target  $R_p(\sqrt{p}) = O(\sqrt{p})$ . The Dodgson identity (Remark 14) restricts local configurations in  $\Gamma_p$ , but by itself controls  $O(H)$  pairs out of  $O(H^2)$ ; upgrading it to a global codegree bound is the key missing step.

## 5.4 Computational evidence

**Proposition 27.** *For all 78,496 primes  $5 \leq p \leq 10^6$ :*

1.  $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$  for all but two primes; the two exceptions ( $p = 11$  and  $p = 3137$ ) have  $Z(p) = 1$  (non-ordinary primes, where  $p \mid a_p(f)$ ).  $\max Z(p) = 12$  at  $p = 159,977$ .
2. The exact histogram is:

| $Z(p)$             | 0     | 1    | 2     | 4    | 6    | 8    | 10   | 12   |
|--------------------|-------|------|-------|------|------|------|------|------|
| count              | 47632 | 2    | 23730 | 6045 | 951  | 123  | 12   | 1    |
| fraction           | .607  | .000 | .302  | .077 | .012 | .002 | .000 | .000 |
| $Poi(\frac{1}{2})$ | .607  | —    | .303  | .076 | .013 | .002 | .000 | .000 |

The mean  $Z(p) = 1.000$ , stable across all ranges ( $\chi^2 = 3.43$  on 5 d.f. against Poisson(1/2) for the pair count).

3. The symmetry  $b_j \equiv b_{p-1-j} \pmod{p}$  holds for all tested primes: zeros come in palindromic pairs  $\{j, p-1-j\}$ , so  $Z(p)$  is even (except at non-ordinary primes). The pair count  $Z(p)/2$  fits Poisson(1/2) to within sampling noise (table above). Proposition 17 proves Poisson(1/2) rigorously in the prime-aspect; the orbit-aspect fit (Remark 18) remains empirical.

*Remark 28* (Mesoscopic root count). At the scale  $H = \lfloor \sqrt{p} \rfloor$ , the total root count  $R_p(H) = \sum_{h=2}^H r_p(h)$  satisfies  $R_p(H)/H \approx 3/2$  for all tested primes, consistent with the Chebotarev prediction from Remark 16:

| $p$    | $H = \lfloor \sqrt{p} \rfloor$ | $R_p(H)$ | $R_p(H)/H$ |
|--------|--------------------------------|----------|------------|
| 503    | 22                             | 35       | 1.59       |
| 2003   | 44                             | 60       | 1.36       |
| 5003   | 70                             | 105      | 1.50       |
| 10007  | 100                            | 154      | 1.54       |
| 20011  | 141                            | 230      | 1.63       |
| 50021  | 223                            | 347      | 1.56       |
| 100003 | 316                            | 460      | 1.46       |

The ratio  $R_p(H)/H$  stabilizes near  $3/2$  (range 1.32–1.75 for  $p > 500$ ), supporting the mesoscopic conjecture in Remark 22.



## 5.5 Hypothesis Z

**Hypothesis 29** (Hypothesis Z). *There exists an absolute constant  $C$  such that  $Z(p) \leq C$  for all primes  $p$ .*

*Remark 30.* The analogous problem for the  $\zeta(2)$  Apéry numbers  $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$  has a fundamentally different structure. Stienstra–Beukers [16] showed that  $A_p$  is related to a CM modular form, yielding  $A_{(p-1)/2} \equiv 0 \pmod{p}$  for all  $p \equiv 3 \pmod{4}$  — a *positive density* of primes with guaranteed zeros. In contrast, the  $\zeta(3)$  Apéry numbers are governed by the non-CM newform  $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$  (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis  $\bar{Z}$  should hold for the  $\zeta(3)$  sequence, even though the pointwise bound  $Z(p) \leq C$  (Hypothesis Z) likely fails.

*Remark 31.* Even the single-row slice  $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$  is connected to the non-ordinary primes of  $f$  via the Beukers congruence  $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$  [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [10]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is  $\sim C_f/p$  (the Sato–Tate distribution on the normalized coefficients  $a_p/p^{3/2}$  admits  $\sim 4p^{1/2}$  multiples of  $p$  in  $[-2p^{3/2}, 2p^{3/2}]$ , each contributing mass  $\sim p^{-3/2}$ ). The expected count up to  $x$  is thus  $C_f \log \log x$  with  $C_f \approx 1$ . Finding 2 non-ordinary primes ( $p = 11, 3137$ ) below  $10^6$  is consistent with  $\log \log(10^6) \approx 2.9$ . This is not a standard Lang–Trotter problem: the target set  $\{a : p \mid a\}$  varies with  $p$ , and no analogue of Elkies’ theorem (infinitely many supersingular primes) is known for weight  $\geq 4$ .

*Remark 32* (Evidence against Hypothesis Z). The  $2 \cdot \text{Poisson}(1/2)$  distribution (Remark 7 below) has unbounded support: extreme-value theory predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ . The current record  $Z(p) = 12$  at  $p = 159,977$  (with record progression 2, 4, 6, 8, 10, 12) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with  $Z(p) = 14$  should appear near  $p \approx 2 \times 10^7$  (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to  $p = 5 \times 10^7$  would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

**Hypothesis 33** (Hypothesis  $\bar{Z}$ ).  $\sum_{p \leq x} Z(p) \ll \pi(x)$ , *equivalently, the average of  $Z(p)$  over primes  $p \leq x$  is bounded.*

*Remark 34.* Hypothesis  $\bar{Z}$  is strictly weaker than Hypothesis Z and consistent with  $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$  (which gives  $\mathbb{E}[Z(p)] = 1$ , independent of  $p$ ). It suffices for the conditional bound  $\log G_n = O(\sqrt{n})$  (Theorem 42).

## 6 The denominator connection lemma

**Lemma 35.** *Let  $p \geq 7$  be prime with  $n/2 < p \leq n$ , and set  $r = n - p$ . Then  $v_p(G_n) = 0$  if and only if  $p \nmid b_r$ .*

*Proof.* Since  $p > n/2$ , the only multiple of  $p$  in  $\{1, \dots, n\}$  is  $p$  itself, so  $v_p(d_n) = 3$ .

**Step 1.**  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $v_p(G_n) = 0$  requires  $v_p(d_n a_n) = 0$ , i.e.,  $v_p(a_n) = -3$ .

**Step 2 (variation of parameters).** From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (12)$$

For  $p > n/2$ , the only  $m \in \{1, \dots, n\}$  with  $p \mid m$  is  $m = p$ .

**Step 3.** By Theorem 4,  $b_p \equiv b_0 b_1 = 5 \pmod{p}$  (since  $p$  has base- $p$  digits  $(0, 1)$ ), and by the symmetry  $b_j \equiv b_{p-1-j}$ ,  $b_{p-1} \equiv b_0 = 1 \pmod{p}$ . Since  $p \geq 7$ , both are nonzero mod  $p$ .

The  $m = p$  term in (12) has  $v_p = -3$ ; all other terms have  $v_p \geq 0$ . Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using  $b_n \equiv 5 b_r \pmod{p}$  (Theorem 4, since  $n = 1 \cdot p + r$ ):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5 b_r = 6 b_r \pmod{p}.$$

Thus  $v_p(a_n) = -3$  iff  $p \nmid b_r$ . □

**Verification.** For all  $n \leq 300$  and primes  $p \in (n/2, n]$  with  $p \geq 7$ :  $v_p(G_n) = 0 \iff p \nmid b_{n-p}$  confirmed in 2101/2101 cases (100%).

**Corollary 36** (Lucas reduction). *For  $p \geq 7$  and  $p \in (n/2, n]$ :  $v_p(G_n) \geq 1$  iff  $p \mid b_{n-p}$ . By the Lucas congruence  $b_n \equiv b_1 b_{n-p} = 5 b_{n-p} \pmod{p}$ , this is equivalent (for  $p \neq 5$ ) to  $p \mid b_n$ . Thus  $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$  equals  $\omega_{(n/2, n]}(b_n) + O(1)$ , the number of distinct prime factors of  $b_n$  lying between  $n/2$  and  $n$ , up to an absolute constant.*

The full conjecture  $G_n = e^{o(n)}$  for all  $n$  is therefore equivalent to the *top-half radical estimate*:

$$\log \text{rad}_{(n/2, n]}(b_n) = o(n). \tag{13}$$

Since  $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52 n$ , the height bound gives  $\omega_{(n/2, n]}(b_n) \leq \log b_n / \log(n/2) = O(n / \log n)$ , precisely one  $o(1)$  factor short of the target. Computation for  $n \leq 200,000$  gives  $B(n) \leq 3$  and  $B(n) = 0$  for 93.7% of values (see Remark 37 for the Poisson analysis).

*Remark 37* (Random-model prediction). Under a random-integer model,  $\Pr(p \mid b_n) \approx 1/p$  for each prime  $p$ , so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\left(\frac{\log n}{\log(n/2)}\right) + O(1/\log n) \approx \frac{\log 2}{\log n} \rightarrow 0.$$

Computation for  $n \leq 200,000$  confirms:  $\mathbb{E}[B] \approx 0.065$ ,  $\max B(n) = 3$ , and the histogram matches  $\text{Poisson}(\log 2 / \log n)$  to high accuracy:  $B(n) = 0$  for 93.7%,  $B(n) = 1$  for 6.0%,  $B(n) = 2$  for 0.20%,  $B(n) = 3$  for 0.006% (the Poisson predictions with  $\lambda = 0.065$  are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio  $\text{Var}(B)/\mathbb{E}[B] = 1.003$  is the exact Poisson signature. By CRT-based approximate independence across primes,  $B(n)$  should follow  $\text{Poisson}(\log 2 / \log n)$  with  $\lambda \rightarrow 0$ . Extreme-value theory then predicts  $\max_{n \leq N} B(n) = O(\log N / \log \log N)$ —trivially  $o(n / \log n)$ —so the random model *strongly* predicts the full conjecture. The obstruction is proving that  $b_n$  behaves “randomly” with respect to its prime factors in  $(n/2, n]$ , despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

## 7 The unconditional density-one theorem

**Theorem 38.** *For a set of positive integers  $n$  of natural density 1,  $G_n = e^{o(n)}$ .*

*Proof.* Fix  $N$  large and consider  $n \in (N, 2N]$ . Write  $B(n) = \#\{\text{bad primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$  for every  $n$ . Each bad prime contributes  $v_p(G_n) \log p \leq 3 \log n$ , so  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ . It suffices to show  $B(n) = o(n/\log n)$  for all but  $o(N)$  values of  $n \in (N, 2N]$ .

For  $p > \sqrt{n}$ ,  $n$  has two base- $p$  digits:  $n = qp + r$  with  $1 \leq q \leq \sqrt{2N}$  and  $0 \leq r < p$ . By Theorem 4,  $p \mid b_n$  iff  $p \mid b_q$  or  $r \in \mathcal{Z}_p$ , where  $\mathcal{Z}_p = \{j < p : p \mid b_j\}$ . A prime  $p$  can be bad for  $n$  only if one of these holds.

**Lower-digit incidence.** For fixed  $p \in (\sqrt{N}, 2N]$ , an interval of  $N$  consecutive integers contains at most  $N \cdot Z(p)/p + Z(p)$  values of  $n$  whose residue  $r = n \bmod p$  lies in  $\mathcal{Z}_p$ . Set  $\rho_p = Z(p)/p$ . The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 9,  $\rho_p \rightarrow 0$ . Since  $\pi(2N) = O(N/\log N)$ :

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence  $R_N = o(N^2/\log N)$ .

**Leading-digit incidence.** For fixed  $q$ ,  $p \mid b_q$  requires  $p$  to be a prime divisor of  $b_q$ , and  $b_q \leq (q+1) \cdot 64^q$  gives at most  $O(q)$  such primes. For each pair  $(q, p)$ , at most  $p$  integers in  $(N, 2N]$  have  $\lfloor n/p \rfloor = q$ . Hence the total leading-digit incidence is

$$L_N \leq \sum_{q=1}^{\lceil \sqrt{2N} \rceil} O(q) \cdot \frac{2N}{q+1} = O(N\sqrt{N}) = o(N^2/\log N).$$

**Conclusion.** Write  $\sum_{n \in (N, 2N]} B(n) \leq R_N + L_N = o(N^2/\log N)$ . For any fixed  $\delta > 0$ , Markov's inequality gives

$$\#\{n \in (N, 2N] : B(n) > \delta n/\log n\} \leq \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so  $B(n) = o(n/\log n)$  for density 1. For all such  $n$ ,  $\log G_n = O(\sqrt{n}) + 3B(n) \log n = O(\sqrt{n}) + o(n) = o(n)$ .  $\square$

**Corollary 39** (Quantitative exceptional set). *For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{2/3}).$$

*Proof.* We make the density-one argument quantitative using the explicit bound  $Z(p) = O(p^{2/3})$  from Proposition 9. For  $n \in (N, 2N]$ ,  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ , so  $\log G_n > \varepsilon n$  requires  $B(n) > \varepsilon n/(7 \log N)$  for  $N$  large.

Partial summation with  $Z(p) \leq Cp^{2/3}$  gives

$$\sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} \leq C \sum_{p \leq 2N} p^{-1/3} = \frac{3}{2} C \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies  $R_N = N \sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$  (the second sum,  $\sum Z(p) \leq C \sum p^{2/3}$ , contributes the same order). The leading-digit incidence  $L_N = O(N^{4/3}/\log N)$  (Proposition 41) is dominated by  $R_N$ . Hence  $\sum_{n \in (N, 2N]} B(n) = O(N^{5/3}/\log N)$ .

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon N / (7 \log N)\} \leq \frac{O(N^{5/3} / \log N)}{\varepsilon N / (7 \log N)} = O(N^{2/3} / \varepsilon).$$

A dyadic decomposition over intervals  $(N/2^{j+1}, N/2^j]$  gives the global bound, since  $\sum_{j \geq 0} (N/2^j)^{2/3} = O(N^{2/3})$ .  $\square$

*Remark 40.* The exponent  $2/3$  in the exceptional set is inherited from the lower-digit incidence  $R_N$ , which uses  $Z(p) = O(p^{2/3})$ . More generally, if  $Z(p) \leq C p^\alpha$  for all primes, then  $R_N = O(N^{1+\alpha} / \log N)$ , while the leading-digit incidence  $L_N = O(N^{4/3} / \log N)$  (Proposition 41) is independent of  $\alpha$ . The exceptional set is therefore  $O(\max(N^\alpha, N^{1/3}) / \varepsilon)$ : for  $\alpha \geq 1/3$  the lower-digit part dominates, and for  $\alpha < 1/3$  the leading-digit part is the bottleneck. Under Hypothesis  $\bar{Z}$  ( $\alpha = 0$ ), the exceptional set is  $O(N^{1/3} / \varepsilon)$ ; Theorem 42 shows the matching density-one bound  $\log G_n = O(\sqrt{n})$ , where  $O(\sqrt{n})$  is now the small-prime floor, not the leading-digit contribution.

## 8 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from  $O(N^{3/2})$  to  $O(N^{4/3} / \log N)$  via a dyadic divisor-counting argument; this is unconditional and also strengthens the conditional theorem below.

**Proposition 41** (Dyadic leading-digit bound). *The leading-digit incidence  $L_N$  (counting pairs  $(n, p)$  with  $n \in (N, 2N]$ ,  $p > \sqrt{n}$  prime, and  $p \mid b_{\lfloor n/p \rfloor}$ ) satisfies*

$$L_N = O\left(\frac{N^{4/3}}{\log N}\right).$$

*Proof.* For dyadic blocks  $P = 2^j$ ,  $Q = 2^k$  with  $Q < P$  and  $PQ \in [N/4, 4N]$ , define

$$D(P, Q) = \#\{(p, q) : P < p \leq 2P, Q < q \leq 2Q, p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Prime counting:*  $D(P, Q) \leq \pi(2P) \leq 2P / \log P$ .
- *Divisor counting:* Since  $\log b_q \leq (q+1) \log 64$ , the number of prime factors of  $b_q$  exceeding  $P$  is at most  $\log b_q / \log P = O(q / \log P)$ . Summing over  $q \in (Q, 2Q]$  gives  $D(P, Q) \leq \sum_q O(q / \log P) = O(Q^2 / \log P)$ .

Hence  $D(P, Q) \leq \min(P, Q^2) / \log P$ .

Each pair  $(p, q)$  in  $D(P, Q)$  contributes to at most  $p \leq 2P$  values of  $n$  with  $\lfloor n/p \rfloor = q$ , so the block's contribution to  $L_N$  is at most  $D(P, Q) \cdot 2P \leq 2 \min(P^2, N^2/P) / \log P$ . The crossover is at  $P_0 = N^{2/3}$ . Summing  $O(\log N)$  dyadic blocks  $P \in [\sqrt{N}, 2N]$ :

$$L_N \leq \sum_{P \leq N^{2/3}} \frac{2P^2}{\log P} + \sum_{P > N^{2/3}} \frac{2N^2}{P \log P} = O\left(\frac{N^{4/3}}{\log N}\right),$$

since both sums are geometric series dominated by the  $P = N^{2/3}$  boundary term.  $\square$

**Theorem 42.** Assume Hypothesis  $\bar{Z}$ . Then for a set of positive integers  $n$  of natural density 1,

$$\log G_n = O(\sqrt{n}).$$

*Proof.* Fix  $N$  and  $\omega(N) \rightarrow \infty$  arbitrarily slowly. As in Theorem 38, decompose  $\log G_n = \sum_p v_p(G_n) \log p$ . The small-prime contribution is  $O(\sqrt{n})$ . Split  $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$ , where  $B_{\text{low}}$  counts bad primes  $p > \sqrt{n}$  with  $n \bmod p \in \mathcal{Z}_p$  and  $B_{\text{lead}}$  counts those with  $\lfloor n/p \rfloor \in \mathcal{Z}_p$ .

**Lower digit.** Under Hypothesis  $\bar{Z}$ , set  $M(t) = \sum_{p \leq t} Z(p) \ll t/\log t$ . Abel summation gives  $\sum_{\sqrt{N} < p \leq 2N} Z(p)/p = O(1)$ , so  $R_N = \sum B_{\text{low}}(n) = O(N)$ . By Markov,  $B_{\text{low}}(n) \leq \omega(N)$  for all but  $O(N/\omega(N)) = o(N)$  values.

**Leading digit.** By Proposition 41,  $L_N = O(N^{4/3}/\log N)$ . By Markov with threshold  $\omega(N) N^{1/3}/\log N$ :

$$\#\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N) N^{1/3}/\log N\} \leq \frac{O(N^{4/3}/\log N)}{\omega(N) N^{1/3}/\log N} = O(N/\omega(N)) = o(N).$$

**Conclusion.** For all remaining  $n$ :

$$\log G_n \leq O(\sqrt{n}) + 3(\omega(N) + \omega(N) N^{1/3}/\log N) \log n.$$

Since  $\omega(N) \cdot N^{1/3} \ll N^{1/2}$  for any  $\omega(N) \rightarrow \infty$  slowly enough ( $\omega \leq N^{1/6}$  suffices), the second term is  $O(N^{1/2}) = O(\sqrt{n})$ .  $\square$

*Remark 43* (The leading-digit bottleneck). The gap between the conditional bound  $O(\sqrt{n})$  (Theorem 42) and the conjectured  $O(\log n)$  comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of  $O(\sqrt{n})$ . Under Hypothesis  $\bar{Z}$ , the big-prime contribution is now absorbed into this floor, thanks to Proposition 41.

Heuristically,  $B_{\text{lead}}(n) = O(1)$  because the leading digit  $\lfloor n/p \rfloor$  lands in  $\mathcal{Z}_p$  with probability  $\leq C/p$  for each prime, giving expected  $B_{\text{lead}} = \sum C/p = O(1)$ . Making this rigorous would require  $D(P, Q) \ll Q/\log P$  (the random-divisibility prediction), improving the proven  $D(P, Q) \leq Q^2/\log P$  by a factor of  $Q$ . The small-prime bound  $O(\sqrt{n})$  itself would need a finer analysis of  $v_p(G_n)$  for small primes.

**Corollary 44** (Conditional exceptional set). Assume Hypothesis  $\bar{Z}$ . For every  $\varepsilon > 0$ ,

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{1/3}).$$

*Proof.* Under Hypothesis  $\bar{Z}$ ,  $R_N = O(N)$ . By Proposition 41,  $L_N = O(N^{4/3}/\log N)$ . The threshold  $B(n) > \varepsilon N/(7 \log N)$  is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7 \log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over  $(N/2^{j+1}, N/2^j]$  gives the global bound.  $\square$

**Corollary 45.** The condition  $Z(p) = o(p)$  is necessary for the conjecture: if  $Z(p) \geq \varepsilon p$  for a positive proportion of primes, double-counting gives  $\log G_n \geq c\varepsilon n$  for a positive proportion of  $n$ . Conversely,  $Z(p) = o(p)$  implies  $G_n = e^{o(n)}$  for density 1 (Theorem 38).

*Proof.* For each prime  $p \in (n/2, n]$  with  $Z(p) \geq \varepsilon p$ , Lemma 35 gives  $v_p(G_n) \geq 1$  whenever  $n-p \in \mathcal{Z}_p$ . Suppose a proportion  $\delta > 0$  of primes satisfy  $Z(p) \geq \varepsilon p$ . Summing over  $n \in (N, 2N]$  and bad primes  $p \in (n/2, n]$ , the total incidence count is  $\geq \delta \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$ . Dividing by  $N$  and applying Markov's inequality: a proportion  $\geq \varepsilon\delta/4$  of integers  $n$  have  $B(n) \geq \varepsilon\delta N/(4 \log N)$ , giving  $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta n$ .  $\square$

*Remark 46* (The pointwise gap). The full conjecture  $G_n = e^{o(n)}$  for *every*  $n$  is strictly stronger than the density-one statement. By Corollary 36, it is equivalent to the top-half radical estimate (13). The height bound gives  $B(n) = O(n/\log n)$ , but for an *adversarial* family  $(Z_p)$  with  $|Z_p| \leq 1$ , one may choose  $Z_p = \{N-p\}$  for primes in  $(N/2, N]$ , producing  $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$ . Thus the needed pointwise bound  $B(n) = o(n/\log n)$  requires arithmetic information specific to the Apéry sequence, not just the cardinalities  $Z(p)$ .

By Corollary 36, every bad prime  $p > n/2$  satisfies  $p \mid b_n$ . Hence the conjecture implies  $\log \text{rad}_{(n/2, n]}(b_n) = o(n)$ : the large prime radical of  $b_n$  must be subexponential, even though  $|b_n| = e^{\Theta(n)}$ .

## 9 Computational verification

We computed  $G_n$  exactly for  $n = 1, \dots, 1000$  using rational arithmetic.

| $n$ range | avg $\log(G_n)/n$ | max $\log(G_n)/n$ |
|-----------|-------------------|-------------------|
| 1–50      | 0.276             | 0.93              |
| 51–100    | 0.108             | 0.26              |
| 101–200   | 0.086             | 0.18              |
| 201–300   | 0.048             | 0.10              |
| 301–400   | 0.038             | 0.08              |
| 401–500   | 0.024             | 0.05              |
| 501–600   | 0.029             | 0.06              |
| 601–700   | 0.017             | 0.04              |
| 701–800   | 0.014             | 0.04              |
| 801–900   | 0.016             | 0.04              |
| 901–1000  | 0.013             | 0.03              |

A power-law fit gives  $\log(G_n)/n \sim 5n^{-0.87}$ , i.e.,  $\log G_n \sim 5n^{0.13}$ , far stronger than the density-one bound  $o(n)$  of Theorem 38.

The exponent 0.13 should be interpreted with caution: if  $\log G_n \sim \kappa \log n$ , the effective log-log slope is  $1/\log n = 0.145$  at  $n = 1000$ , close to the fitted 0.13. A random-gcd model—assigning each prime  $p \leq n$  an independent probability  $\sim 1/p$  of dividing  $G_n$ —predicts  $\mathbb{E}[\log G_n] \sim \log n$  by the weighted Mertens theorem, with standard deviation  $\sim (\log n)/\sqrt{2}$  of the same order. The data thus cannot yet distinguish  $\kappa \log n$  from  $n^{0.13}$ ; the logarithmic prediction is arithmetically more natural.

## 10 Remarks

1. The Beukers supercongruence  $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$  [4] provides tower rigidity: for each fixed prime,  $v_p(b_n)$  is controlled by the base- $p$  digits.
2. For the gap-2 polynomial  $C_2(m) = P(m+1)$ , the factorization  $P(x) = (2x+1)(17x^2+17x+5)$  shows that gap-2 pairs can only occur at  $m = (p-3)/2$  (when  $2(m+1)+1 = 2m+3 \equiv 0$ ) or at roots of the quadratic  $17x^2+17x+5$  (discriminant  $-51$ ; exists over  $\mathbb{F}_p$  iff  $(\frac{-51}{p}) = 1$ ). Empirically, all observed gap-2 pairs arise from the linear factor at  $m = (p-3)/2$ .
3. For  $1 \leq j \leq p-2$ , the Apéry number  $b_j \bmod p$  admits a character-sum representation as a Greene [8] Gaussian hypergeometric  ${}_4F_3$  over  $\mathbb{F}_p$  with parameter  $\chi = \omega^j$  (Teichmüller character). However, the Weil bound controls the archimedean size  $|b_j|_\infty = O(p^{3/2})$  but does

not bound  $Z(p)$ , since zeros of  $b_j$  modulo  $p$  are a  $p$ -adic event. The gap polynomial argument above is specifically adapted to this  $p$ -adic zero-count problem.

4. Malik–Straub [11] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [14] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our  $Z(p)$  data for the  $\zeta(3)$  sequence agrees with their empirical findings.
5. In the standard toric coordinate, the truncated polynomial  $H_p(t) = \sum_{j=0}^{p-1} b_j t^j \in \mathbb{F}_p[t]$  is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in  $\mathbb{F}_p$  are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic  $Z(p)$  counts vanishing *coefficients* of  $H_p$ , equivalently the vanishing finite Mellin coefficients  $b_j = -\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j}$  for  $1 \leq j \leq p-2$ . Thus  $Z(p)$  is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent  $F$ -crystal theory do not by themselves bound  $Z(p)$ ; they control the root locus, not the Fourier support. A bounded-conductor realization of  $H_p$  as an  $\ell$ -adic trace function could connect to the Forey–Fresán–Kowalski framework [7], but equidistribution of complex traces does not directly control exact zeros modulo  $p$ .
6. **Palindromic symmetry.** For  $0 \leq j \leq p-1$  and  $0 \leq k \leq p-1-j$ , the identity  $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$  holds (each of the  $k$  falling factors  $p-1-j-i \equiv -(j+1+i)$ ). When  $k \leq j$ ,  $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$ ; when  $k > j$ , Lucas’s theorem gives  $\binom{p-1-j+k}{k} \equiv 0$  (since  $p-1-j+k \geq p$  and  $\binom{k-j-1}{k} = 0$ ). Therefore each term of  $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$  with  $k > j$  vanishes, and for  $k \leq j$  the  $(-1)^{2k}$  signs cancel, giving  $b_{p-1-j} \equiv b_j \pmod{p}$ . The extra terms of  $b_j$  when  $j > (p-1)/2$  (i.e.,  $k > p-1-j$ ) likewise vanish:  $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$  since  $j+k-p < k$ . Equivalently,  $H_p(t)$  is palindromic:  $t^{p-1} H_p(1/t) \equiv H_p(t)$ . This explains the observed pairing: whenever  $b_j \equiv 0$ , also  $b_{p-1-j} \equiv 0$ , so zeros of  $b_j$  come in pairs  $\{j, p-1-j\}$  (except at the self-paired center  $j = (p-1)/2$ , which vanishes only at non-ordinary primes). The “pair count”  $K(p) = Z(p)/2$  (or  $(Z(p)-1)/2$  at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 27).
7. **Sym<sup>2</sup> squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [16]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [6] proved the mod- $p$  squareness rigorously: for every prime  $p \geq 5$ , setting  $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$ , there exists a unique (up to sign) polynomial  $B_p \in \mathbb{F}_p[t]$  such that

$$H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where  $\Delta(t) = t^2 - 34t + 1$  is the conifold discriminant and  $B_p$  is squarefree and coprime to  $\Delta$ . The proof uses the explicit identity  $F(t(x)) = (1+x)h(x)^2$  between the Apéry and Franel generating functions, the  $p$ -Lucas property of both sequences, and a quadratic-descent argument through the covering  $\mathbb{F}_p(x)/\mathbb{F}_p(t)$  where  $t = x(1-8x)/(1+x)$ . The deck-transformation character  $(-6/p)$  determines whether  $\varepsilon_p = 0$  or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \quad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently,  $\varepsilon_p = 0$  iff  $p \equiv 1, 5, 7, 11 \pmod{24}$ . These two cases occur with equal density 1/2 by Dirichlet’s theorem.

Splitting the mean  $Z(p)$  by residue class modulo 8 (and modulo 24) yields no significant difference (mean  $\approx 1.00$  in each class,  $n \geq 2000$  primes per class up to  $10^5$ ); the squareness structure does not directly influence  $Z(p)$ , consistent with  $Z(p)$  counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure:  $b_j$  is (up to the  $\Delta^{\varepsilon_p}$  correction) the  $j$ -th coefficient of  $B_p^2$ , i.e.,  $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$  where  $\beta_k$  denotes the coefficients of  $B_p$ . The vanishing  $b_j \equiv 0$  thus requires cancellation across  $\sim p/2$  convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree  $(p-1)/2$  over  $\mathbb{F}_p$ , for  $p \in \{101, 503, 1009, 5003\}$ ) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean  $\approx 1$ , variance  $\approx 1$ , total variation distance to Poisson(1) equal to 0.018). However, the observed  $Z(p)$  distribution over 667 primes  $p \leq 5000$  is *not* Poisson(1) (TV = 0.43); instead it matches  $2 \cdot \text{Poisson}(1/2)$  with TV = 0.023. The explanation is the palindromic pairing: since  $b_{p-1-j} \equiv b_j$ , zeros come in pairs  $\{j, p-1-j\}$  and  $Z(p)$  is almost always even. Denoting by  $Z_{\text{pair}}(p)$  the number of zero pairs, the data show  $Z_{\text{pair}} \sim \text{Poisson}(1/2)$ , so  $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$ . Thus squareness explains the  $O(1)$  mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$  via extreme-value theory, strongly suggesting that  $Z(p)$  is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of  $B_p$  imposes a bounded *average* (Hypothesis  $\bar{Z}$ ) remains the central open question.

8. **Toric framework.** The Laurent polynomial  $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$  on  $(\mathbb{F}_p^\times)^4$  has Newton polytope  $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$ , a 4-dimensional pyramid over the regular octahedron in  $\mathbb{R}^3$ . Its  $f$ -vector is  $(7, 18, 20, 9)$ , its normalized volume is  $4! \cdot \text{vol}(\Delta) = 8$ , and it is Kouchnirenko nondegenerate for every prime  $p \geq 5$ : for each of the 27 nonempty faces  $\sigma$  not containing the origin, the toric critical system  $\{x_i \partial f_\sigma / \partial x_i = 0\}$  has no solution in  $(\mathbb{F}_p^\times)^4$ . For the Adolphson–Sperber twisted Hodge polygon with  $w$ -Kummer twist  $\chi_w = \omega^{-j}$  (character parameter  $a = j/(p-1)$ ), the slopes are  $a, a+1, a+2, a+3$  with multiplicities  $1, 3, 3, 1$ , confirming that the first slope equals  $a$  for every  $a \in \{k/100 : 0 \leq k \leq 99\}$ . By Newton-above-Hodge (Adolphson–Sperber [2]), the  $p$ -adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The  $Z(p)$  bound thus reduces to proving ordinarity for all but  $O(1)$  twist parameters  $j$ .
9. **Branch obstruction.** The  $p-1$  Kummer twists  $\chi^{-j}$  ( $j = 0, \dots, p-2$ ) are distinct characters of conductor  $p$ . In Iwasawa-theoretic coordinates  $\chi^j = \omega^a \langle \cdot \rangle^s$  with  $a = j \pmod{p-1}$ , the horizontal family  $\{j : 0 \leq j < p-1\}$  hits *each* of the  $p-1$  branches  $\omega^a$  exactly once. Weierstrass preparation and bounded- $\lambda$  rigidity control zeros *within one branch* (the vertical direction  $j, j + (p-1), j + (p-1)p, \dots$ , where Dwork congruences live) but not *across* branches. Thus there is no  $p$ -adic analytic function of one variable whose zero count majorizes  $Z(p)$ . Any “rigidity” proof of Hypothesis  $\bar{Z}$  would need a mechanism not supplied by existing Iwasawa–Dwork theory.
10. Any estimate  $\log G_n = o(n)$ —whether  $O(\log n)$ ,  $O(\sqrt{n})$ , or  $O(n^{0.13})$ —yields  $\gamma = 0$  in the exponent  $G_n = e^{\gamma n + o(n)}$  and therefore does not improve the Rhin–Viola irrationality measure  $\mu(\zeta(3)) \leq 5.513$  [15], which depends only on the exponential growth rate  $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$ . An improvement would require an exponentially large common divisor, which is not predicted by any current model.



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